

Point Patterns I

HES 505 Fall 2025: Session 15

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Update on Assignments

- Updated (clarified) process:
 - You submit (commit) your first version
 - I'll post the key
 - You update your assignment highlighting any areas you still feel unsure about
 - I'll annotate your repo with feedback

Update on Assignments

- Key to Assignments 1 and 2 posted by Oct 15.
- Revisions Due by Nov 1.
- Key to Assignment 3 posted by Oct 27.
- Revision due Nov 12

Point Patterns

Objectives

- Define a point process
- Define Complete Spatial Randomness
- Distinguish First and Second-Order CSR
- Introduce density-based methods for describing point patterns

What is a point pattern?

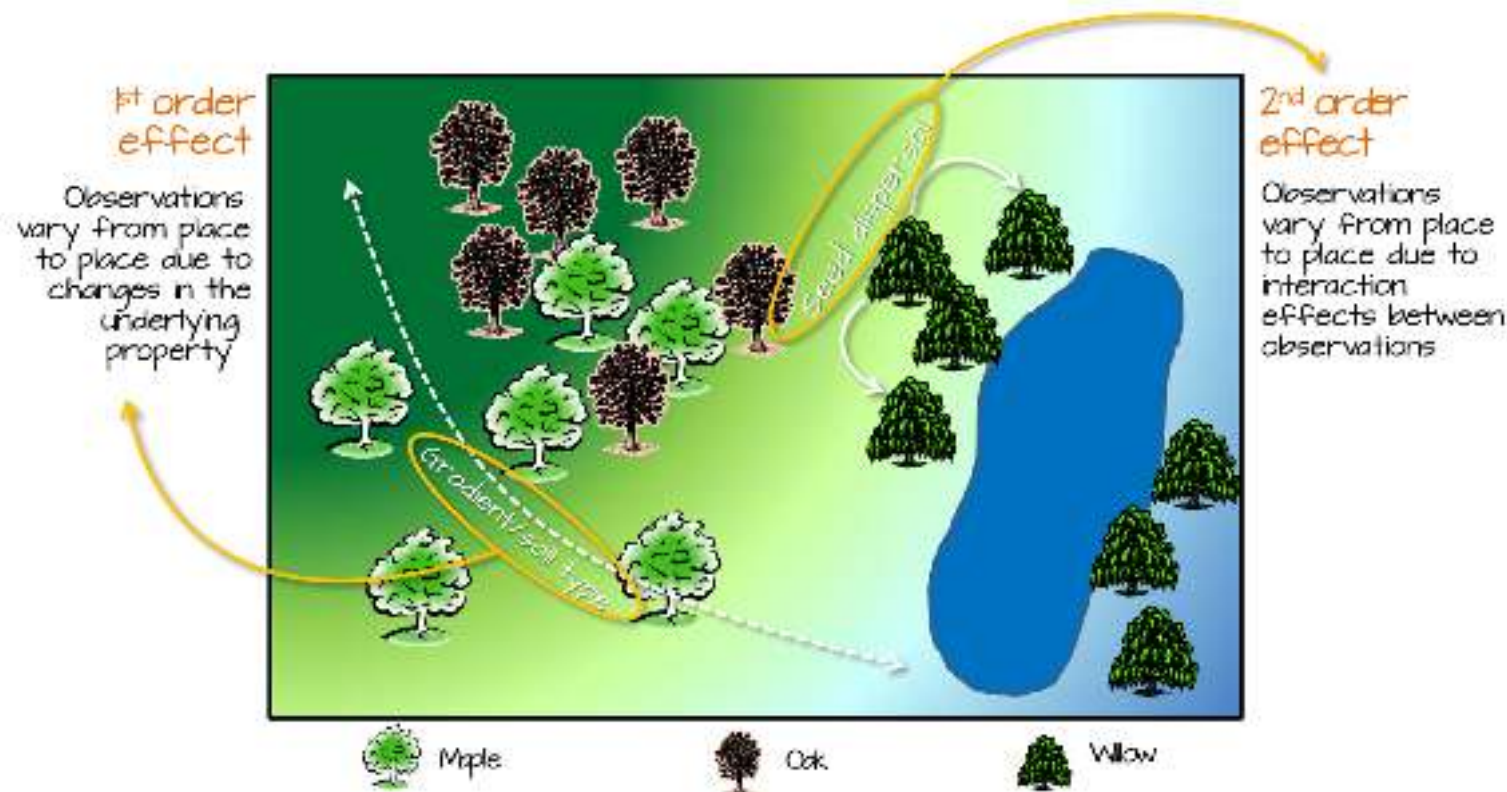
- *Point pattern*: A **set** of **events** within a study region (i.e., a *window*) generated by a random process
- **Set**: A collection of mathematical **events**
- **Events**: The existence of a point object of the type we are interested in at a particular location in the study region
- A *marked point pattern* refers to a point pattern where the events have additional descriptors

Some notation:

- $\mathcal{S}$: refers to the entire set
- $\mathbf{s}_i$ denotes the vector of data describing point s_i in set $\mathcal{S}$
- $\#(\mathcal{S} \cap A)$ refers to the number of points in $\mathcal{S}$ within study area A

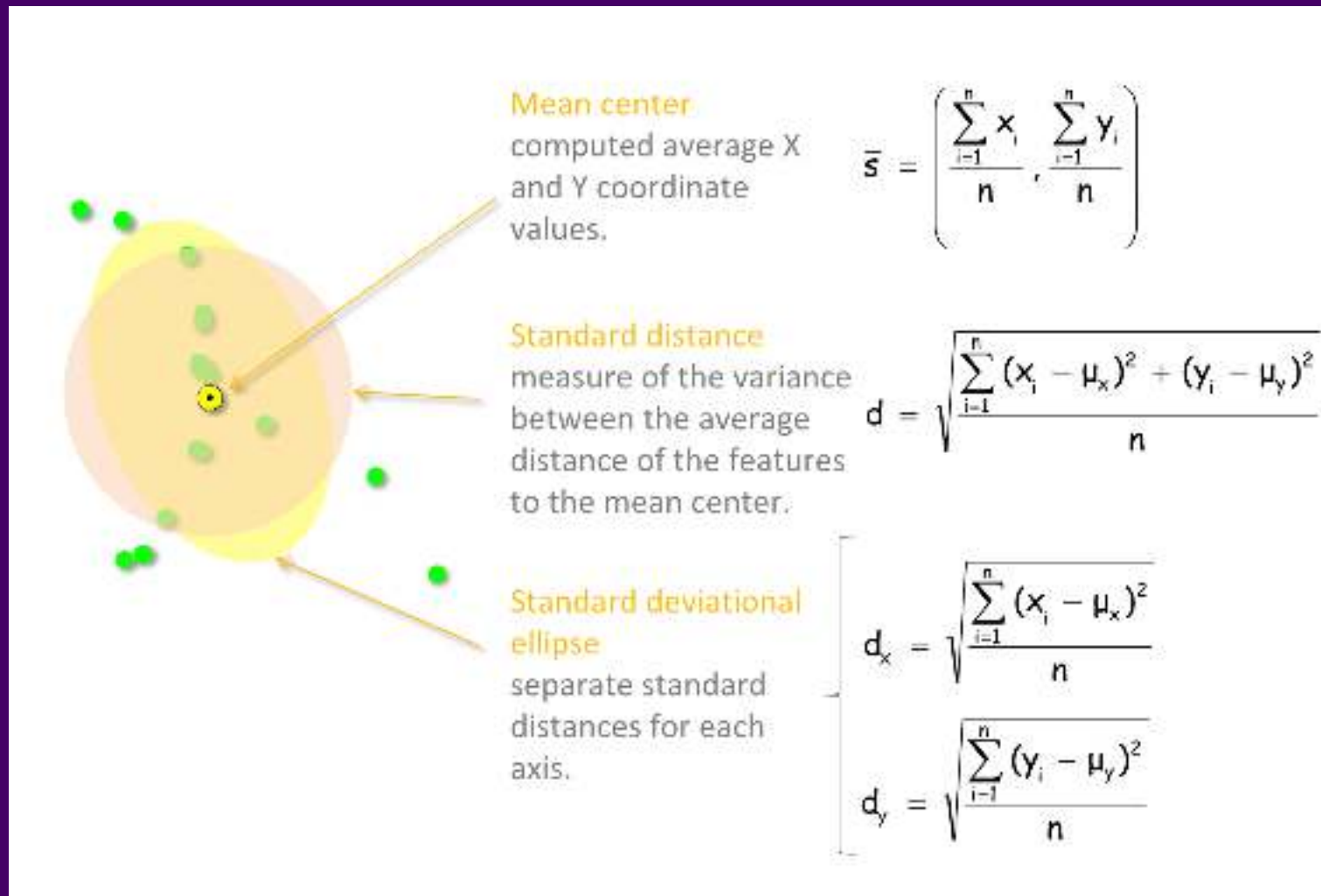
Requirements for a set to be considered a point pattern

Describing Point Patterns



- *First order* effects reflect variation in **intensity** due to variation in the ‘attractiveness’ of locations (density)
- *Second order* effects reflect variation in **intensity** due to the presence of points themselves (distance)

Basic Descriptions: Centrography



Analyzing Point Patterns

3 Flavors of Point Pattern Analysis

- Modeling random processes means we are interested in probability densities of the points (first-order; density)
- Also interested in how the presence of some events affects the probability of other events (second-order; distance)
- Finally interested in how the attributes of an event affect location (marked)
- Need to introduce a few new packages (**spatstat** and **gstat**)

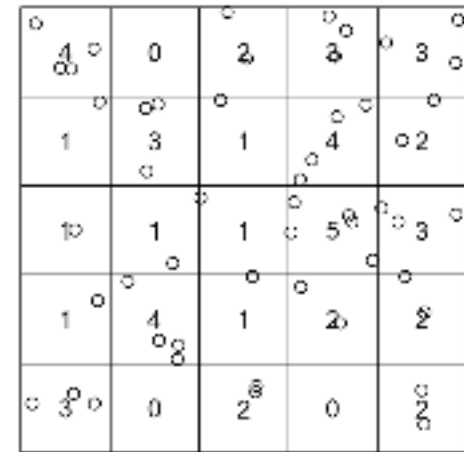
First Order Analysis: CSR

- Homogeneous Poisson Process:
 - All events have an equal likelihood of occurring anywhere
 - Point locations are independent (no attraction/repulsion)
 - Given a rate (λ) we can estimate the expected number of points

Density based methods

- The overall *intensity* of a point pattern is a crude density estimate
- Local density = quadrat counts

$$\hat{\lambda} = \frac{\#(S \cap A)}{a}$$



Quadrat Counts and CSR

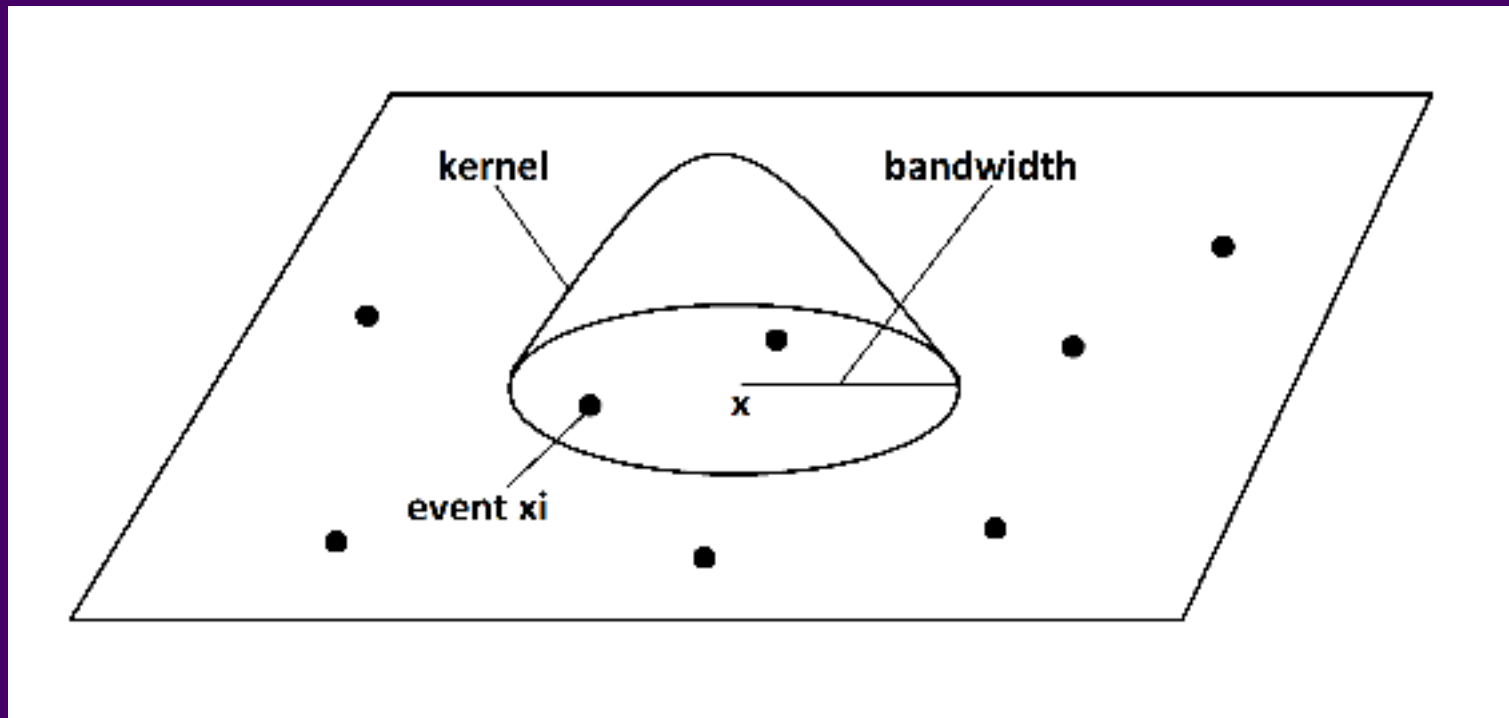
- Observed vs. Expected
- χ^2 test

Kernel Density Estimates (KDE)

$$\hat{f}(x) = \frac{1}{nh_x h_y} \sum_{i=1}^n k\left(\frac{x-x_i}{h_x}, \frac{y-y_i}{h_y}\right)$$

- Assume each location in $(\mathbf{s}_i)$ drawn from unknown distribution
- Distribution has probability density $(f(\mathbf{x}))$
- Estimate $(f(\mathbf{x}))$ by averaging probability “bumps” around each location
- Need different object types for most operations in R (`as.ppp`)

Kernel Density Estimates (KDE)



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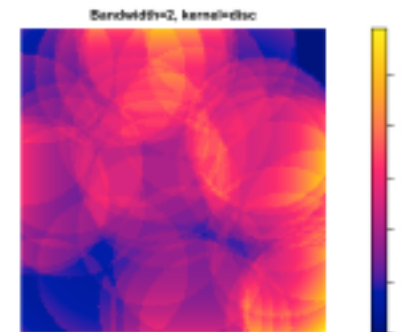
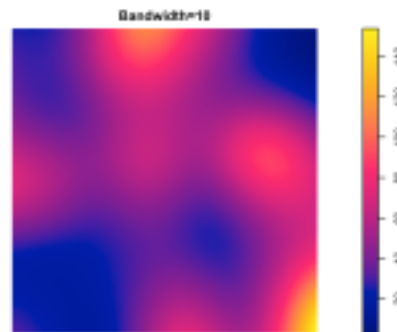
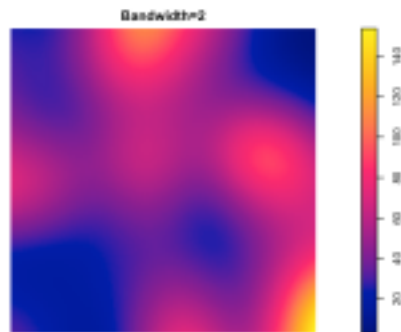
$$\hat{f}(x) = \frac{1}{nh_x h_y} \sum_{i=1}^n k\left(\frac{x-x_i}{h_x}, \frac{y-y_i}{h_y}\right)$$

- (h) is the bandwidth and (k) is the kernel
- We can use `stats::density` to explore
- **kernel**: defines the shape, size, and weight assigned to observations in the window
- **bandwidth** often assigned based on distance from the window center

Bandwidth Matters (more than kernel shape)

- Look at `?density` to see all the kernel options!

```
1 x <- rpoispp(lambda = 50)
2 K1 <- density(x, bw=2)
3 K2 <- density(x, bw=10)
4 K3 <- density(x, bw=2, kernel="disc")
```



Choosing bandwidths and kernels

- Small values for $\hat{h}$ give 'spiky' densities
- Large values for $\hat{h}$ smooth much more
- Some kernels have optimal bandwidth detection

